

Fixed-node diffusion Monte Carlo computations for closed-shell jellium spheres

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Fixed-node diffusion Monte Carlo computations are used to determine the ground state energy and electron density for jellium spheres with up to $N=106$ electrons and background densities corresponding to the electron gas parameter $1 \leq r_s \leq 5.62$. We analyze the density and size dependence of the surface energy, and we extrapolate our data to the thermodynamic limit. The results agree well with the predictions of density functional computations using the local density approximation. In the case of $N=20$, we extend our computation to higher densities and identify a transition between atomic- and jelliumlike nodal structures occurring at the background density corresponding to $r_s=0.13$. In this case the local density approximation is unable to reproduce the changes in the correlation energy due to the discontinuous transition in the ground state nodal structure. We discuss the relevance of our results for nonlocal approximations to density functional theory.

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I. INTRODUCTION

Approximate theories for the electronic structure of condensed matter systems rely, to a large extent, on our knowledge of the homogeneous electron gas, whose properties have been investigated extensively by analytical¹ and computational methods.² The simplest extensions to inhomogeneous systems, i.e., planar jellium slabs and jellium spheres, have been investigated far less by methods of comparable accuracy, despite their obvious interest as benchmarks for approximate theories³ and their ability to model inhomogeneous valence electron systems decoupled from deeper levels. The presence of such levels may obscure the comparison between theory and experiments in real systems. Semianalytical computations for jellium slabs have been performed by Krotscheck, Kohn, and Qian⁴ within the Fermi hypernetted chain approximation, while a limited set of densities and system sizes have been investigated by Monte Carlo computations.⁵ To the best of our knowledge, the only quantum Monte Carlo results for spherical jellium spheres have been reported in Ref. 6, for systems with up to $N=20$ electrons.

In this paper, fixed-node diffusion Monte Carlo⁷ (FN-DMC) computations for jellium spheres are performed, with the aim of extending the range of densities and sizes for which accurate data for the total energy and the electron density are available. We focus on system sizes ($N=2, 8, 18, 20, 34, 40, 58, 92$, and 106) that are known to correspond to closed-shell configurations with a spherical ground state, and we explore a set of densities ($r_s=1, 2, 3.25, 4$, and 5.62) covering the valence density range of simple (i.e., *sp* bonded) systems. The total energy and (especially) the electron density profile predicted by density functional computations using the local density approximation (DFT-LDA) agree well with the results of our simulations.

Besides providing benchmark data to test approximate theories, our computations allow us to investigate the interplay of shell effects and correlation in finite (i.e., zero dimensional) systems. In this respect, the $N=20$ electron system is particularly interesting: as discussed in detail in Sec. IV, by

continuously changing the background density, it is possible to transform this jellium sphere into the (closed-shell and spherical) calcium atom, which has a different electronic configuration. This is a discontinuous transition, accompanied by precursor effects in the correlation energy. Density functional computations with the local density approximation correctly locate the transition density, but fail to reproduce crucial qualitative features of the transition, related to the discontinuous change in the nodal structure of the ground state wave function.

Before describing our computation in detail, we point out two traditional reasons for interest in jellium spheres, with a closer relation with experiments. First of all, as shown by recent reviews of the vast literature on this subject (e.g., Refs. 8–10), jellium spheres have been used extensively to model alkali metal clusters,¹¹ and have provided the basis for the interpretation of the shell effects revealed by early mass spectroscopy measurements.¹² Moreover, finite jellium systems have long been used to model alkali metal surfaces,^{13,14} and jellium spheres offer the opportunity to investigate the curvature dependence of the surface energy in simple metals.

II. THE MODEL AND THE SIMULATION METHOD

The model we adopt is the spherical version of the well known jellium model, in which N interacting electrons move in the electrostatic potential $V_{ext}(r)$ of a structureless positive background. The system is globally neutral, so that the radius of the background is given by $R_B=N^{1/3}r_s$, and its density is

$$\rho_B(r) = \begin{cases} \frac{3}{4\pi r_s^3}, & r \leq R_B \\ 0, & r > R_B. \end{cases} \quad (1)$$

The electron gas parameter r_s is introduced in order to allow an easy comparison of the background density with the valence density of simple *sp* electron systems.

The Hamiltonian of the system is (atomic Hartree units are used throughout)

$$H = -\frac{1}{2} \sum_i^N \nabla^2 + \sum_{i=1}^N V_{\text{ext}}(|\mathbf{r}_i|) + \frac{1}{2} \sum_{i \neq j}^N \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + E_{\text{self}}, \quad (2)$$

where $\{\mathbf{r}_i, i=1, N\}$ denote the electron coordinates, $V_{\text{ext}}(r)$ is given by

$$V_{\text{ext}}(r) = \begin{cases} -\frac{1}{2} \frac{N}{R_B} \left(3 - \frac{r^2}{R_B^2} \right), & r < R_B \\ -\frac{N}{r}, & r \geq R_B, \end{cases} \quad (3)$$

and E_{self} is the Coulomb self-energy of the background,

$$E_{\text{self}} = \frac{3}{5} \frac{N^{5/3}}{r_s}. \quad (4)$$

This last contribution is introduced in order to recover, in the $N \rightarrow \infty$ limit, a total energy consistent with the results of previous computations for the homogeneous electron gas. This Hamiltonian has been extensively used to model alkali metal clusters, and both the ground and excited state properties have been investigated by approximate methods for a wide range of sizes and densities (see Refs. 8 and 10 for reviews).

These systems show a pronounced shell structure, i.e., the appearance of stability peaks at sizes corresponding to the filling of orbital momentum levels. This structure is reminiscent of atomic structure, but the external potential is different and the filling follows different orderings in atoms ($1s, 2s, 2p, 3s, \dots$) and jellium ($1s, 1p, 1d, 2s, \dots$), at least for background densities in the valence charge range. In both cases, the shell closing sizes correspond to the energetically most stable structures (and, therefore, for alkali metal clusters they correspond to abundance peaks in mass spectra), and to spherical ground state electron distributions. For these reasons, we focus on the values $N=2, 8, 18, 20, 34, 40, 58, 92, 106$, corresponding to the filling of the levels $1s^2 | 1p^6 | 1d^{10} | 2s^2 | 1f^{14} | 2p^6 | 1g^{18} | 2d^{10} | 1h^{22} | 3s^2 | 2f^{14}$ (vertical lines identify the different sizes we consider).

The computation follows closely the method described in Ref. 6, and we report here only the basic equations used in the present implementation. The simulation is based on a many-body wave function $\Psi_T(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$ optimized by a variational Monte Carlo (VMC) computation. We assume a wave function consisting of the product of spin-up (D_+) and spin-down (D_-) Slater determinants and a Jastrow factor that is symmetric with respect to electron permutations:

$$\begin{aligned} \Psi_T(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \\ = D_+(\mathbf{r}_1, \dots, \mathbf{r}_{N/2}) D_-(\mathbf{r}_{N/2+1}, \dots, \mathbf{r}_N) J(\mathbf{r}_1, \dots, \mathbf{r}_N). \end{aligned} \quad (5)$$

The determinants are built from single-particle orbitals given by the Kohn-Sham eigenstates computed within the local density approximation.¹⁵ The Jastrow factor is assumed to be the exponential of a many-body ‘‘potential’’ $U(\mathbf{r}_1, \dots, \mathbf{r}_N)$,

which is the sum of a one-body contribution V_1 , a two-body contribution V_2 , and a many-body (multipolar) contribution V_M :

$$U(\mathbf{r}_1, \dots, \mathbf{r}_N) = \sum_{i=1}^N V_1(\mathbf{r}_i) + \frac{1}{2} \sum_{i \neq j}^N V_2(\mathbf{r}_i, \mathbf{r}_j) + V_M(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (6)$$

The single-body contribution is given by

$$V_1(|\mathbf{r}_i|) = \sum_{n=1}^6 \alpha_n j_0 \left(\frac{n \pi r_i}{\alpha_0 R_B} \right), \quad (7)$$

where j_0 is the spherical Bessel function of order zero, and $\alpha_0, \dots, \alpha_6$ are free parameters. The two-body term is

$$V_2(\mathbf{r}_i, \mathbf{r}_j) = \frac{1}{2} \left(\frac{a_{ij} r_{ij} + b_{ij} r_{ij}^2 + c_{ij} r_{ij}^3}{1 + d_{ij}(|\mathbf{r}_i|) r_{ij} + e_{ij} r_{ij}^2 + f_{ij} r_{ij}^3} + \frac{a_{ij} r_{ij} + b_{ij} r_{ij}^2 + c_{ij} r_{ij}^3}{1 + d_{ij}(|\mathbf{r}_j|) r_{ij} + e_{ij} r_{ij}^2 + f_{ij} r_{ij}^3} \right), \quad (8)$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, the index ij on the coefficients indicates that they depend on the relative spin (parallel or antiparallel) of electrons i and j , and the dependence of d_{ij} on the electrons’ position models the dependence of the correlation hole size on the local density of the system: in the high density regions close to the system center of mass, correlation is shorter range than in the tail of the electron distribution. This effect is translated into the functional form

$$d_{ij}(r) = d_{ij}^0 + d_{ij}^1 \arctan \left(\frac{r^2 - R_B^2}{2 \Delta R_B} \right). \quad (9)$$

The multipolar contribution is given by

$$V_M(\mathbf{r}_1, \dots, \mathbf{r}_N) = -\beta \mathbf{P}^2 - \gamma \mathbf{P}_s^2 \quad (10)$$

where $\mathbf{P}$ and $\mathbf{P}_s$ are the instantaneous number and spin dependent dipole moments:

$$\mathbf{P} = \sum_{i=1}^N \mathbf{r}_i, \quad (11)$$

$$\mathbf{P}_s = \sum_{i=1}^{N/2} \mathbf{r}_i - \sum_{i=N/2+1}^N \mathbf{r}_i. \quad (12)$$

The bias in the wave function introduced by the multipolar term makes configurations with large fluctuations of $\mathbf{P}$ and $\mathbf{P}_s$ unlikely. The essential idea behind the functional form for the Jastrow factor is to bracket its behavior between the parts determined by few-body (V_1, V_2) interactions and the many-body correlations. The parameters are varied to optimize the variance σ_E of the local energy $E_{\text{loc}} = [H \Psi_T] / \Psi_T$, following the prescriptions of Ref. 17. The total energy and variance of the local energy obtained at the end of our VMC optimization are given in Table I.

We emphasize that the multipolar term in the Jastrow potential gives a sizable contribution to the reduction of both

the energy and its variance, especially for the lowest sizes and highest densities. On the other hand, introducing additional degrees of freedom in the determinants or using a linear combination of determinants (that is, performing a limited configuration interaction computation) improves neither the energy nor σ_E .

Table I reports also E_{tot}^{HF} (and its variance σ_{HF}), which is computed as the expectation value of the Hamiltonian on the product of the spin-up and spin-down Slater determinants, and which we identify with the Hartree-Fock (HF) energy of our systems. This identification is not exact, because the orbitals entering the determinants have been computed by the DFT-LDA, but it is generally accepted that they differ little from the Hartree-Fock orbitals. The effect on the energy is even smaller, since the linear error in the orbitals becomes a quadratic error for the expectation value of the Hamiltonian. The validity of this approximation has been tested by introducing and optimizing parameters in the single-electron orbitals entering the Slater determinants, and we found that this optimization does not significantly improve the energy or its variance.¹⁸ We therefore use E_{HF} of Table I as a reference to estimate the correlation energy recovered by our Monte Carlo computations. The optimized wave function is used to drive the FN-DMC stage, in which we follow the method introduced in Ref. 19, including all the improvements discussed in Ref. 20.

A population of 5000 walkers is propagated in imaginary time by applying the operator $\exp[-\tau(H-E_D)]$, where E_D is a trial energy that ensures the long term stability of the population size. The time step for this propagation is gauged by the bulk plasma frequency ω_P (we adopt $dt=1/250\omega_P$), and we achieved in all cases an acceptance ratio higher than 0.998. The accuracy of our results with respect to the finite time step error was demonstrated by repeating several computations with a time step five times longer. We found differences in expectation values for the total energy that were smaller than the statistical error bar. Each population has been equilibrated for at least 700 steps, before accumulating statistics for at least a further 500 steps. In a few cases (for the smaller systems) the simulation was continued for several thousand additional steps, to check the convergence of our results with respect to the length of the imaginary time diffusion process.

The trial energy E_D for the diffusion process differs by less than the statistical error bar from the time average $\langle E_L \rangle$ of the local energy $[H\Psi_T]/\Psi_T$ on the stationary population of walkers. In the following, we report $\langle E_L \rangle$ (i.e., the “mixed” estimator) as the FN-DMC total energy of the simulated systems.

Quantities different from the total energy (like the density and the radial distribution functions) are given by the “extrapolated” estimator introduced in Ref. 21.

III. FIXED-NODE DIFFUSION MONTE CARLO RESULTS

The total and correlation energies (per electron) for the various sizes and background densities considered in the present study are reported in Table II. For $N=20$ the results are consistent with those of Ref. 6, with the exception of

TABLE I. Total energy per electron (eV) of spherical jellium systems computed by the Hartree-Fock (E_{tot}^{HF}) and variational Monte Carlo (E_{tot}^{VMC}) methods. The corresponding standard deviations of the local energy are given (in eV) by σ^{HF} and σ^{VMC} . The VMC correlation energy per electron E_c^{VMC} (eV) is given in the last column. To reduce the round-off error in the computation of the correlation energy, all the tables report one digit more than consistent with the statistical error bar.

N	E_{tot}^{VMC}	σ^{VMC}	E_{tot}^{HF}	σ^{HF}	E_c^{VMC}
$r_s = 1$					
2	5.0861	0.378	5.5689	5.299	0.4828
8	8.4296	0.593	9.2467	3.750	0.8172
18	10.6803	0.534	11.6716	2.885	0.991
20	10.2709	0.513	11.2700	2.791	0.999
34	11.8559	0.445	12.9347	2.284	1.079
40	11.4223	0.417	12.5271	2.186	1.105
58	12.5124	0.389	13.6583	1.875	1.145
92	12.9156	0.321	14.1125	1.550	1.197
106	12.8678	0.295	14.0970	1.428	1.229
$r_s = 2$					
2	-0.7123	0.162	-0.2600	3.338	0.4523
8	-0.3887	0.353	0.2913	2.169	0.6800
18	-0.1278	0.264	0.6714	1.588	0.7992
20	-0.2338	0.255	0.5681	1.565	0.8019
34	-0.0559	0.212	0.8126	1.238	0.8685
40	-0.1209	0.200	0.7686	1.156	0.8896
58	-0.0386	0.177	0.8664	0.991	0.9049
92	-0.0422	0.145	0.9028	0.804	0.9449
106	0.0049	0.136	0.9727	0.973	0.9678
$r_s = 3.25$					
2	-1.6703	0.106	-1.2616	2.406	0.4087
8	-1.7220	0.181	-1.1276	1.442	0.5944
18	-1.7082	0.156	-1.0363	1.030	0.6719
20	-1.7321	0.158	-1.0551	0.992	0.6770
34	-1.7416	0.124	-1.0177	0.796	0.7240
40	-1.7430	0.117	-1.0033	0.730	0.7397
58	-1.7773	0.0987	-1.0277	0.623	0.7496
92	-1.7991	0.0821	-1.0152	0.508	0.7839
106	-1.7614	0.0801	-0.9623	0.476	0.7991
$r_s = 4$					
2	-1.7431	0.117	-1.3530	2.156	0.3901
8	-1.8128	0.200	-1.2704	1.202	0.5423
18	-1.8379	0.131	-1.2200	0.855	0.6179
20	-1.8522	0.124	-1.2235	0.822	0.6286
34	-1.8826	0.0963	-1.2242	0.652	0.6584
40	-1.8741	0.0911	-1.1918	0.602	0.6823
58	-1.9168	0.0786	-1.2275	0.516	0.6893
92	-1.9347	0.0698	-1.2282	0.413	0.7065
106	-1.9038	0.0635	-1.1761	0.391	0.7777
$r_s = 5.62$					
2	-1.6482	0.0328	-1.2966	1.567	0.3516
8	-1.7192	0.1246	-1.2456	0.897	0.4920
18	-1.7506	0.0911	-1.2235	0.626	0.5270
20	-1.7521	0.0914	-1.2196	0.589	0.5325
34	-1.7940	0.0697	-1.2331	0.470	0.5610
40	-1.7848	0.0629	-1.2033	0.441	0.5815
58	-1.8258	0.0518	-1.2378	0.371	0.5880
92	-1.8433	0.0417	-1.2429	0.302	0.6004
106	-1.8231	0.0416	-1.2042	0.280	0.6189

TABLE II. Total energy (eV) and correlation energy (eV) per electron of spherical jellium systems from fixed-node diffusion Monte Carlo computations. The error bar accounts only for the statistical uncertainty. The last column (E_{tot}^{LDA}) gives the DFT-LDA total energy per electron (eV).

N	E_{tot}^{DMC}	E_c^{DMC}	E_{tot}^{LDA}
$r_s = 1$			
2	5.0810 ± 0.005	0.488	5.4841
8	8.4046 ± 0.008	0.842	8.5671
18	10.6410 ± 0.008	1.031	10.6949
20	10.2331 ± 0.0075	1.037	10.3132
34	11.8080 ± 0.0043	1.127	11.8272
40	11.3766 ± 0.0052	1.151	11.4100
58	12.4543 ± 0.0039	1.204	12.4531
92	12.8601 ± 0.0035	1.252	12.8454
106	12.8184 ± 0.0043	1.279	12.7965
$r_s = 2$			
2	-0.7147 ± 0.0023	0.4548	-0.5188
8	-0.4181 ± 0.0049	0.7094	-0.3580
18	-0.1561 ± 0.0038	0.8274	-0.1446
20	-0.2623 ± 0.0044	0.8303	-0.2492
34	-0.0874 ± 0.0013	0.8999	-0.0933
40	-0.1537 ± 0.0031	0.9224	-0.1644
58	-0.0780 ± 0.0026	0.9443	-0.0957
92	-0.0778 ± 0.0020	0.9806	-0.1009
106	-0.0289 ± 0.0019	1.0016	-0.0599
$r_s = 3.25$			
2	-1.6716 ± 0.0015	0.4100	-1.5803
8	-1.7379 ± 0.0027	0.6103	-1.7213
18	-1.7306 ± 0.0023	0.6943	-1.7376
20	-1.7615 ± 0.0024	0.7065	-1.7687
34	-1.7668 ± 0.0012	0.7491	-1.7807
40	-1.7656 ± 0.00097	0.7624	-1.7861
58	-1.8032 ± 0.0016	0.7754	-1.8216
92	-1.8266 ± 0.0014	0.8115	-1.8489
106	-1.7891 ± 0.0011	0.8268	-1.8153
$r_s = 4$			
2	-1.7440 ± 0.0016	0.3910	-1.6820
8	-1.8349 ± 0.0030	0.5645	-1.8336
18	-1.8624 ± 0.0020	0.6424	-1.8732
20	-1.8784 ± 0.0017	0.6549	-1.8899
34	-1.9035 ± 0.0014	0.6793	-1.9197
40	-1.8972 ± 0.0017	0.7054	-1.9158
58	-1.9390 ± 0.0010	0.7115	-1.9582
92	-1.9626 ± 0.0010	0.7343	-1.9836
106	-1.9326 ± 0.0010	0.7565	-1.9561
$r_s = 5.62$			
2	-1.6485 ± 0.0004	0.3519	-1.6230
8	-1.7377 ± 0.0019	0.4921	-1.7467
18	-1.7758 ± 0.0013	0.5522	-1.7911
20	-1.7804 ± 0.0018	0.5608	-1.7959
34	-1.8126 ± 0.0009	0.5796	-1.8301
40	-1.8026 ± 0.0008	0.5993	-1.8215
58	-1.8419 ± 0.0008	0.6041	-1.8594
92	-1.8605 ± 0.0005	0.6176	-1.8786
106	-1.8417 ± 0.0006	0.6375	-1.8605

those for $r_s = 3.25$.²² From the same table, the good quality of the VMC wave functions is also apparent; they recover at least 95% of the correlation energy estimated at the FN-DMC level, thus enhancing our confidence in the reliability

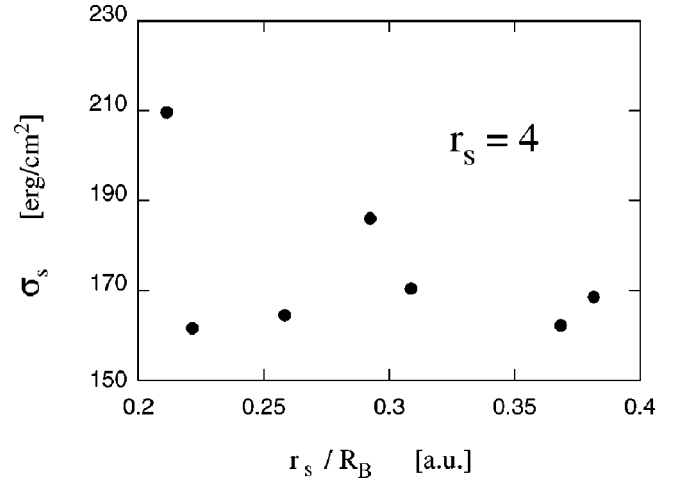


FIG. 1. Surface energies in closed-shell jellium spheres as a function of the inverse of the background radius $R_B = N^{1/3}r_s$.

of the FN-DMC step, and in the insensitivity of the results to time step, choice of the estimators, etc.

Surface energies have been computed as differences between the total energies in Table II and the total energy of the same number of electrons in the homogeneous electron gas, obtained by adding to the Hartree-Fock energy the correlation energy per electron given by the interpolation formula of Ref. 23. This energy difference is then divided by $4\pi R_B^2$ to give the surface energies E^{surf} shown in Fig. 1 for $r_s = 4$. The size dependence of the surface energy displays large and irregular oscillations, which prevent a simple extrapolation to the thermodynamic limit, in order to estimate the surface energy of planar jellium slabs. These oscillations arise mainly from the angular dependence of the kinetic energy, as we showed by analyzing the different contributions to the total energy. Even in simulations for much larger spherical jellium systems, the extrapolation of surface energies to the $1/R_B \rightarrow 0$ limit is subject to large uncertainties, since the accurate computation of E^{surf} for large sizes depends on the exact cancellation of bulk energy contributions. We investigate below different extrapolation schemes based on the analysis of the size dependence of the correlation energy.

Correlation energies per electron for clusters of different size at $r_s = 1$ and $r_s = 4$ are reported in Fig. 2 as a function of the inverse of the background radius R_B , and a linear interpolation (including only the sizes $N \geq 18$) already provides an excellent fit to the FN-DMC data. This regularity is somewhat unexpected, because different sizes correspond to the closing of shells with different angular character, for which correlation effects could be different. The extrapolation of the fit to the thermodynamic limit gives an estimate for the correlation energy consistent with that reported in Ref. 23 for the homogeneous electron gas. Moreover, the linear coefficient of the interpolation gives the correlation contribution to the planar jellium surface energy directly. This can easily be seen by assuming that the correlation energy per electron $e_c(r_s, N)$ has the analytic expansion

$$e_c(r_s, N) = e_c^{heg}(r_s) + \frac{\delta_1(r_s)}{N^{1/3}r_s} + o\left(\frac{1}{[N^{1/3}r_s]^2}\right), \quad (13)$$

where the superscript “*heg*” identifies the different quantities computed in the thermodynamic limit (i.e., for the homogeneous electron gas). The exact and Hartree-Fock total energies for the N electron system are then related by

$$E_{tot}(r_s, N) = E_{HF}(r_s, N) + N \left[e_c^{heg}(r_s) + \frac{\delta_1(r_s)}{N^{1/3}r_s} + o\left(\frac{1}{[N^{1/3}r_s]^2}\right) \right] \quad (14)$$

and the surface energy $E^{surf}(r_s)$ is obtained from

$$\begin{aligned} E^{surf}(r_s) &= \lim_{N \rightarrow \infty} \frac{E_{tot}(r_s, N) - N e_{tot}^{heg}}{4\pi[N^{1/3}r_s]^2} \\ &= \lim_{N \rightarrow \infty} \left\{ \frac{E_{HF}(r_s, N) - N e_{HF}^{heg}(r_s)}{4\pi[N^{1/3}r_s]^2} + \frac{N}{4\pi[N^{1/3}r_s]^2} \right. \\ &\quad \times \left. \left[e_c^{heg}(r_s) + \frac{\delta_1(r_s)}{N^{1/3}r_s} + o\left(\frac{1}{[N^{1/3}r_s]^2}\right) \right] \right\}, \quad (15) \end{aligned}$$

giving

$$E^{surf}(r_s) = E_{HF}^{surf}(r_s) + \delta_1(r_s)/4\pi r_s^3, \quad (16)$$

where $E_{HF}^{surf}(r_s)$ is the Hartree-Fock value for the planar surface energy. None of the quantities entering Eq. (16) depends on the exact cancellation of energies estimated for finite and infinite systems.

These results suggest that a reliable extrapolation to the thermodynamic limit of the total or surface energy obtained for finite systems can be achieved by first isolating the correlation contribution, and then adding its extrapolated value to the Hartree-Fock estimates computed for extended systems (see, for example, Ref. 25).

The $\delta_1(r_s)$ obtained by the linear fit of the computed $e_c(r_s, N)$ (for $N \geq 18$) are listed in Table III, while the corresponding correlation contributions to the surface energy are compared in Fig. 3 with the results provided by a FN-DMC study of planar jellium slabs.^{5(b)} The reasons for the differences between our extrapolated data and those of Ref. 5(b) are still unclear, and we are studying this point further. We emphasize that our estimates for $E^{surf}(r_s)$, if confirmed by FN-DMC computations for planar jellium slabs, would reduce the disagreement between the results provided by recent DFT computations³ and the values predicted by wave function-based methods.^{4,5}

As mentioned in the Introduction, the major aim of quantum Monte Carlo computations for model systems like the jellium spheres is to provide *nearly exact* data in order to test the accuracy of approximate methods. Here we discuss a comparison with the results obtained within the local density approximation of density functional theory,¹⁵ whose assumption of a slow spatial variation of the electron density has often been considered adequate for alkali metals.²⁶ The DFT-

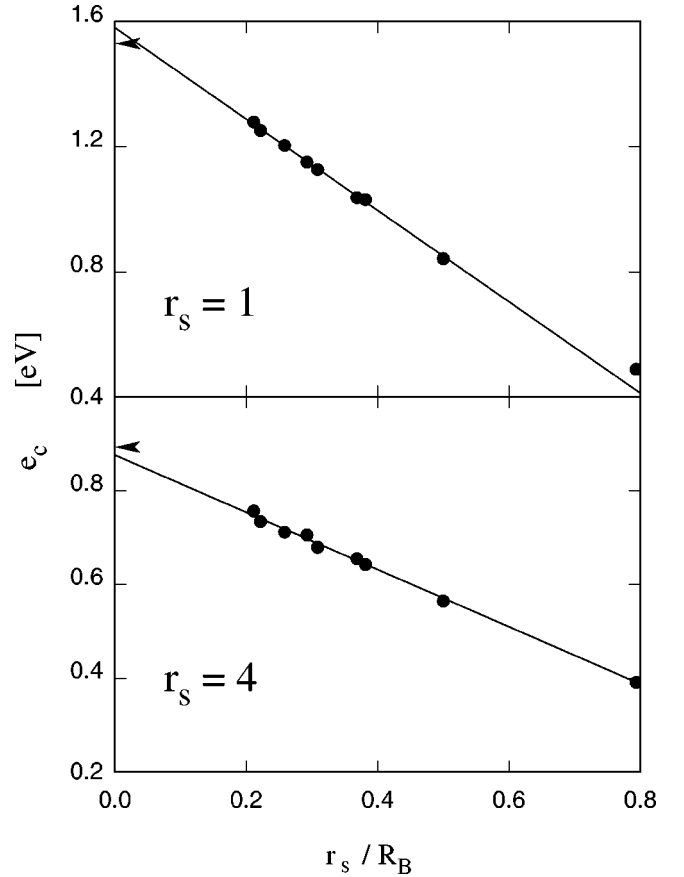


FIG. 2. Dots: correlation energy per electron in closed-shell jellium spheres as a function of the inverse of the background radius $R_B = N^{1/3}r_s$. The value for $\lim R_B \rightarrow \infty$, indicated by the horizontal arrow, is from the homogeneous electron gas computation in Ref. 24. The full line is a linear interpolation of the jellium sphere data.

LDA total energies, reported in Table II, display a good agreement with our FN-DMC results for all but the smallest system size ($N=2$). Not surprisingly, the relative error increases in going to lower densities, corresponding to more strongly interacting systems.

The deviation between the two sets of results does not have a definite sign: E_{tot}^{LDA} is above E_{tot}^{DMC} for the smallest sizes and higher densities, while the reverse is true for larger or more dilute systems. This is in contrast with the well known result $E_{tot}^{LDA} > E_{tot}^{DMC}$ for all atoms, in which the residual electron Hartree self-energy is the most important source of error for the LDA energy and potential.^{16,27}

Electron density profiles from our FN-DMC computations are reported in Fig. 4, where they are compared with the

TABLE III. Results of a linear fit for the size dependence of the correlation energy per electron $e_c(r_s, N) = e_c(r_s, N) + \delta_1(r_s)/r_s/N^{1/3}$. $e_c(r_s, N)$ and $\delta_1(r_s)/r_s$ are given in eV. The fit has been performed for sizes $N \geq 18$.

r_s	1	2	3.25	4	5.62
$\tilde{e}_c(r_s)$	1.58	1.21	0.98	0.88	0.72
$\delta_1(r_s)$	-1.460	-1.015	-0.736	-0.610	-0.450

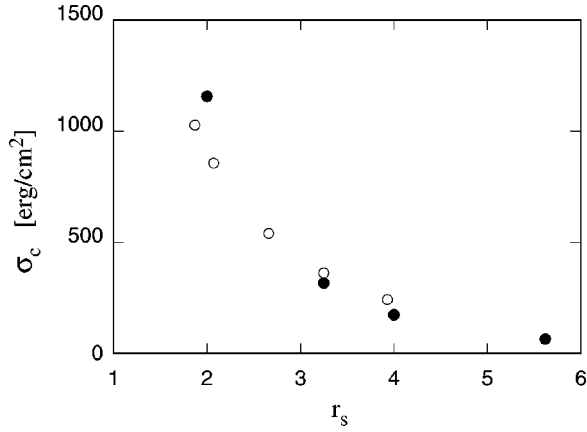


FIG. 3. Correlation energy contribution to the planar jellium surface energy as a function of r_s . Full dots: results of the present computation, based on the extrapolation defined in Eq. (16). Empty circles: from Ref. 5(b).

LDA results. The agreement between the two sets of data is remarkably good in this case as well.

In principle, the availability of a virtually exact density distribution opens the possibility of computing the “exact” Kohn-Sham potential by inverting the Kohn-Sham equations. Unfortunately, this inversion is ill conditioned, and even the small statistical noise apparent in Fig. 4 introduces substan-

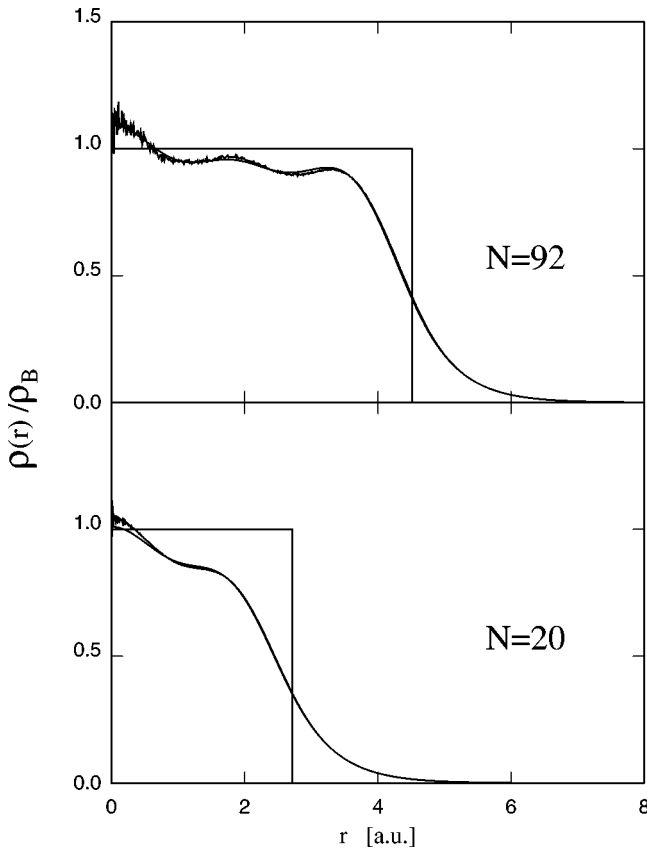


FIG. 4. Electron density profile for two different sizes at $r_s = 1$. Full line: DFT-LDA computation. Dashed line: FN-DMC computation.

tial uncertainties. In particular, it has been impossible to verify by purely computational means that, for the exact Kohn-Sham potential, the eigenvalue of the highest occupied orbital equals the system ionization potential.²⁸ A second attempt to determine this eigenvalue by analyzing the decay of the density at large distances has been frustrated by the observation that $\rho(r)$ acquires its asymptotic behavior, dominated by a single exponential decay, so far from the edge of the background that the accumulation of sufficient statistics would require a prohibitively long simulation.

The effect of inhomogeneity on the local coordination of electrons has been investigated by computing the pair correlation function $g_2(\mathbf{r}_1, \mathbf{r}_2)$ around an electron located at the edge of the background [$\mathbf{r}_1$ has Cartesian coordinates $(0, 0, R_B)$]. We distinguish the cases of pairs with parallel and antiparallel spins. These are denoted $g_2^l(\mathbf{r}_1, \mathbf{r}_2)$ and $g_2^u(\mathbf{r}_1, \mathbf{r}_2)$, respectively, the superscripts l and u being shorthand for *like* and *unlike* spin pairs, respectively. Below we give the definitions for the case of antiparallel spins. The parallel-spin definitions are similar, with trivial differences in the factors accounting for the number of equivalent pairs in the system. For a system with an equal number ($N/2$) of spin-up and spin-down electrons, $g_2^u(\mathbf{r}_1, \mathbf{r}_2)$ is defined as

$$\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)g_2^u(\mathbf{r}_1, \mathbf{r}_2) = \frac{N(N-1)}{4} \frac{\int d\mathbf{r}_3 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2}{\int d\mathbf{r}_1 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2}, \quad (17)$$

where the electrons at $\mathbf{r}_1$ and at $\mathbf{r}_2$ have opposite spin. Moreover, since

$$\rho(\mathbf{r}_1) = \frac{N}{2} \frac{\int d\mathbf{r}_2 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2}{\int d\mathbf{r}_1 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2}, \quad (18)$$

we obtain

$$g_2^u(\mathbf{r}_1, \mathbf{r}_2) = \frac{(N-1)}{2\rho(\mathbf{r}_2)} \frac{\int d\mathbf{r}_3 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2}{\int d\mathbf{r}_2 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2}. \quad (19)$$

This last quantity can be computed easily by keeping an electron fixed at $\mathbf{r}_1$, performing the usual FN-DMC (or VMC) sampling of all the other electron coordinates, and recording the (like- or unlike-spin) density around the fixed electron. Because of our choice of $\mathbf{r}_1 = (0, 0, R_B)$, $g_2^u(\mathbf{r}_1, \mathbf{r}_2)$ has azimuthal symmetry, and it can be expanded in the sum of the $m=0$ spherical harmonic components centered at $\mathbf{r}_1$:

$$g_2^u(\mathbf{r}_1, \mathbf{r}_2) = \tilde{g}_{00}^u(\mathbf{r}_1, |\mathbf{r}_1 - \mathbf{r}_2|) Y_{00}(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2) + \tilde{g}_{10}^u(\mathbf{r}_1, |\mathbf{r}_1 - \mathbf{r}_2|) Y_{10}(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2) + \cdots \quad (20)$$

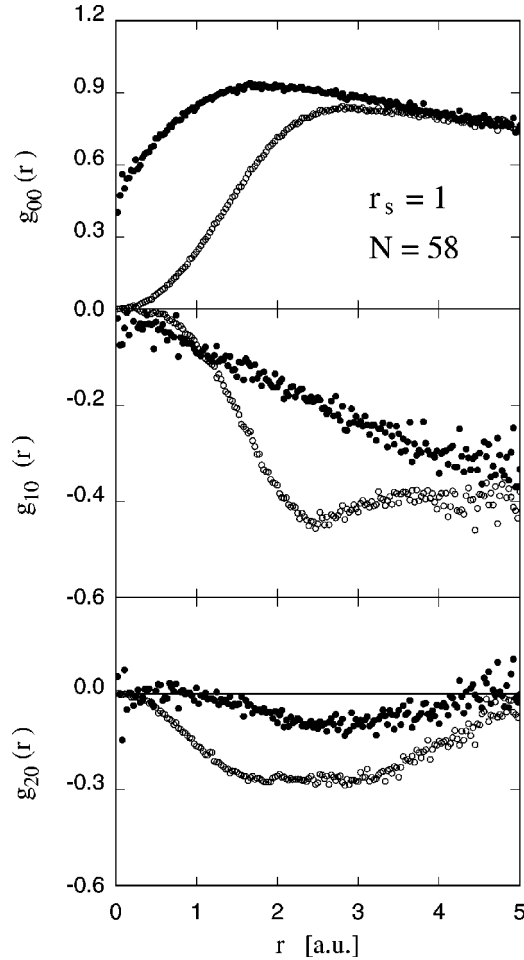


FIG. 5. Lowest order multipolar components of the pair correlation function (see text). The three functions displayed are defined as $g_{00}(r) = \tilde{g}_{00}(r)/\sqrt{4\pi}$, $g_{10}(r) = \tilde{g}_{10}(r)\sqrt{3/4\pi}$, $g_{20}(r) = \tilde{g}_{20}(r)\sqrt{5/4\pi}$, where the tilde functions are defined in Eq. (20). In homogeneous systems, $g_{00}(r)$ tends to 1 at large distances. Solid dots, unlike-spin pairs; empty circles, like-spin pairs.

The first three lowest order components are displayed in Fig. 5 for $r_s = 1$ and $N = 58$. It is apparent that the spherical average does not tend to unity at large distances,²⁹ and the range of the lowest order multipolar components is not short. These features are shared by all the systems we analyzed. With decreasing density of the background, the pair correlation function displays changes similar to those seen in homogeneous systems.² On the other hand, the results for different sizes show differences that are not systematic, and probably depend on the angular character of the orbitals contributing to the electron density at $\mathbf{r}_1$. Despite the difference in dimensionality, our results for the pair distribution function are qualitatively similar to those of Ref. 30 for a circular (i.e., two-dimensional) quantum dot, suggesting that the features described above might be a general property for small inhomogeneous systems, in which single-electron orbitals extend over the entire system, and electron interactions cannot be screened effectively by the (highly discretized) population of unoccupied states.

Our pair correlation functions are also qualitatively simi-

lar to the results of Ref. 5(b), although a precise comparison is prevented by the difference in the simulated systems and by uncertainties in interpreting the contour plots in Ref. 5(b).

IV. THE ATOM-JELLIUM TRANSITION FOR THE $N=20$ ELECTRON SYSTEM

In this section we analyze in detail the properties of a spherical 20 electron system as a function of the background density and radius. In the density range considered in the previous section, the ground state has a closed-shell $1s^2 1p^6 1d^{10} 2s^2$ electronic configuration. However, if we imagine increasing the background density and decreasing its radius while keeping constant the integrated positive charge, we obtain the limiting case of a pointlike positive charge corresponding to the atomic case. The $N=20$ jellium sphere must transform into the calcium atom, which has a different $(1s^2 2s^2 2p^6 3s^2 3p^6 4s^2)$ electronic configuration. We expect that this transition will be reflected here by a similar transformation in the determinant part of the optimized trial wave function, from $\Psi_j(1s^2 1p^6 1d^{10} 2s^2)$ at low background density to $\Psi_a(1s^2 2s^2 2p^6 3s^2 3p^6 4s^2)$ at high background density, drastically changing the nodal structure of the ground state wave function. The analysis of this transition and the comparison with its DFT-LDA description provide additional insight into both the success and the well known failures of simple density functional approximations.

Because of the fair agreement of energies and densities computed by FN-DMC and DFT-LDA, we use the latter for a first exploration of the density range ($r_s < 1$) in which the transition between the jelliumlike ($1s^2 1p^6 1d^{10} 2s^2$) and the atomiclike ($1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$) configurations has to take place. The LDA total energy for both configurations is reported in Fig. 6, where we see a transition at $r_s \sim 0.13$. This density is three orders of magnitude higher than typical valence electron densities, and the system total energy ($E_{tot} = -310$ hartrees) is still far above that of the Ca atom ($E_{tot} = -677$ hartrees). Close to the transition, it is possible to lower the value of the DFT-LDA functional below the heavy line in Fig. 6 by resorting to fractional occupation numbers mixing the two configurations. These mixed states are excluded from our discussion because (as we show below) there is no simple correspondence with the configuration interaction that can be included in our MC computations.

In the DFT-LDA framework, if we restrict ourselves to single-determinant Kohn-Sham wave functions, the situation is fairly simple: for $r_s \leq 0.13$ the ground state adopts the atomic configuration and the jellium configuration represents an excited state, while the reverse is true for $r_s > 0.13$. The change from one to the other is abrupt, with a discontinuous jump in the energy derivative with respect to the r_s parameter. The transition, therefore, is analogous to first order transitions in statistical mechanics. Moreover, the analysis of the different contributions to the total energy shows that the exchange-correlation part is fairly continuous across the transition, while the slopes of the kinetic and Coulomb parts display significant changes.

We then analyzed by VMC and FN-DMC methods the

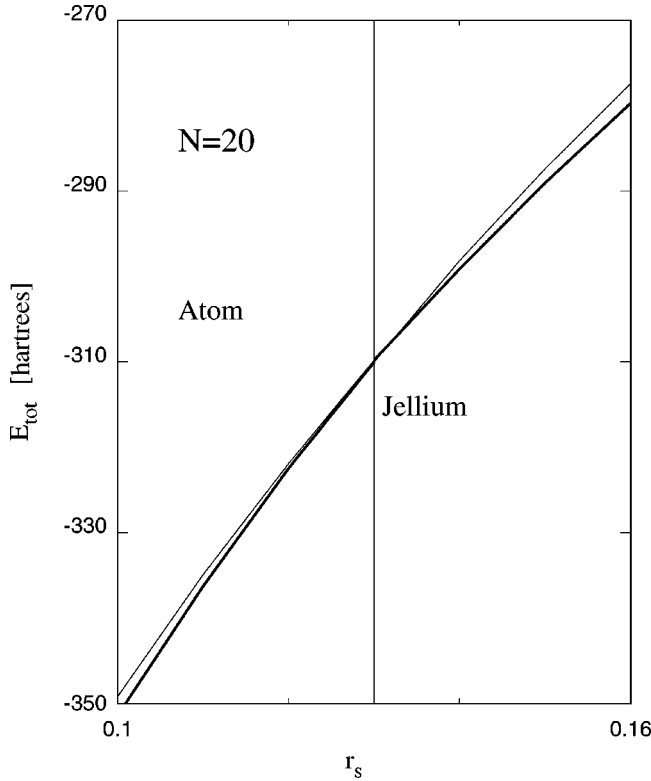


FIG. 6. Total energy as a function of the background r_s parameter for two different configurations (atom and jellium, see text). The heavy line identifies the ground state energy, that corresponds to the atom configuration for $r_s \leq 0.13$, and to the jellium configuration for $r_s > 0.13$.

densities $r_s = 0.1, 0.13, 0.15, 0.18$, and 0.20 .³¹ Far from the crossing point, the two wave functions Ψ_a (atomic) and Ψ_j (jellium) (Ref. 32) are separated by a large energy difference, and perturbation theory shows that mixing them in a configuration interaction scheme will not lead to a significant energy gain. Close to the crossing point, one might expect that configuration mixing would transform the abrupt transition into a smooth “avoided crossing” between the two states. Instead, the large difference in the two angular nodal structures (the two wave functions have only the first two s and the first p orbitals in common) makes the matrix element $\langle \Psi_j | H | \Psi_a \rangle$ vanishingly small, preventing an effective interaction of the two states. For each background density, therefore, the optimal determinant part is either atomic- or jelliumlike and the transition is abrupt in VMC or FN-DMC as in the DFT-LDA. Moreover, the density at which the transition takes place is almost the same in these different computational schemes. At the crossing point, the same external potential corresponds to two different ground state densities, an unusual situation that is not, however, forbidden.³³

First, we focus our attention on correlation energy, because this is the contribution that always needs to be approximated by practical DFT schemes. The behavior of correlation close to the transition point is illustrated in Fig. 7, which shows the correlation energy per electron computed by DFT-LDA, VMC, and FN-DMC methods. The FN-DMC correlation energy at the transition differs by as much as

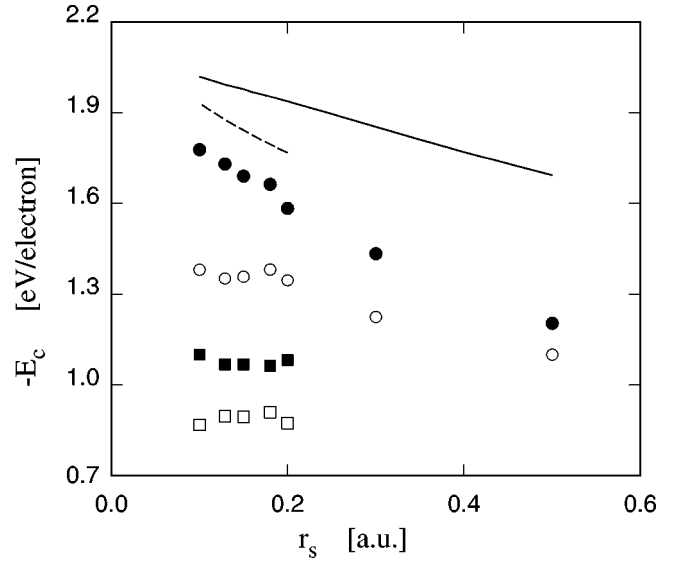


FIG. 7. Correlation energy per electron close to the jellium to atom transition (see text). Solid and empty circles, E_c for the jellium configuration by the FN-DMC and VMC methods, respectively; filled and empty squares, E_c for the atom configuration by the FN-DMC and VMC methods, respectively; full line, DFT-LDA results for the jellium configuration; dashed line, DFT-LDA results for the atom configuration.

~60% for the two configurations. Furthermore, while E_c is almost constant in the atom configuration, it displays a strong density dependence in the jellium configuration. The VMC results do not follow the rapid rise of the jellium E_c for $r_s \leq 0.2$, suggesting that the excess correlation at these densities is of the nondynamic variety,³⁴ i.e., it is due to the interaction of the ground state with a multitude of excited configurations.³⁵ In particular, if we interpret correlation energy as the result of the perturbation represented by $H - H_{HF}$ on the uncorrelated (HF) ground state, we see that, by decreasing r_s , the jellium state is increasingly responsive to the perturbation, i.e., it manifests increasing symptoms of instability. With the transition, correlation is discontinuously reduced, corresponding to an increased rigidity of the HF ground state.

The DFT-LDA results for the correlation energy do not reproduce the simulation picture: as well as the well known overestimation of E_c ,¹⁶ the DFT-LDA correlation energy differs by at most 10% in the jellium and atom configurations, while the density dependence of E_c is similar for the two configurations, and close to the FN-DMC result for jellium. In Fig. 8 the DFT-LDA total energies are compared with the FN-DMC results for the two different configurations: it is apparent that the DFT-LDA estimate is always very close to the jellium exact result, while it is unable to reproduce the total energy of the atom configuration.

We emphasize that, for each density and for each configuration, the electron density given by the FN-DMC simulations does not deviate significantly from the DFT-LDA result, and the jellium and atom density distributions differ significantly both at the DFT-LDA and at the VMC or FN-DMC levels of approximation. The implications of these ob-

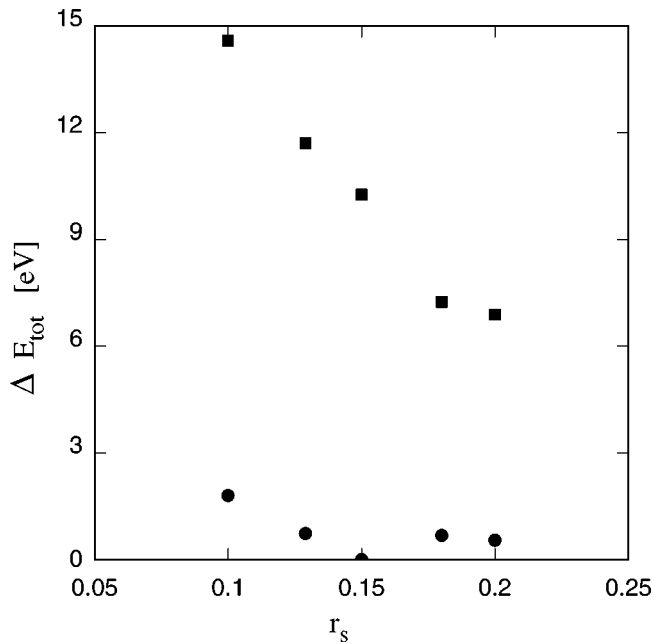


FIG. 8. Difference in the total energy $\Delta E_{\text{tot}} = E_{\text{tot}}^{\text{LDA}} - E_{\text{tot}}^{\text{DMC}}$ for 20 electron spheres in the jellium configuration (dots) and in the atom configuration (circles).

servations for the reliability of approximate DFT schemes are briefly discussed in the following section.

V. CONCLUSIONS

By the fixed-node diffusion Monte Carlo technique we have analyzed the total energy and electron density for jellium spheres with up to $N=106$ electrons. Our results provide accurate data for the surface energy of these finite systems and an estimate for the energy of planar jellium surfaces. More importantly, our results provide an accurate benchmark to test the reliability of standard density functional computations. The comparison with LDA shows that this simple approximation provides a fairly accurate description of total energies for finite systems whose density is in the range $1 \leq r_s \leq 5.62$. In particular, for these densities (typical of valence electrons), the deviation between the DFT-LDA and FN-DMC total energies can be either negative or positive, vanishing in the middle of the size and density range analyzed here. This is at variance with results for atoms, for which the DFT-LDA overestimates the exact total energy due to a large self-interaction contribution in the Hartree term.

In the $N=20$ case, computations have been extended to higher densities, and the results show a transition in the electronic configuration from jelliumlike to atomiclike occurring

at $r_s=0.13$. The transition appears to be discontinuous and is accompanied on the low density side by an anomalous increase of correlation energy, pointing to an increasing instability of the single-electron orbital configuration.

The transition marks a discontinuity point also for the performances of the LDA approximation: while in the low density (jellium) region the LDA provides an excellent description of the total energy, the quality of this approximation degrades drastically at the transition, and becomes progressively worse at higher densities. In other words, the DFT-LDA is able to reproduce the total energy of systems that are adiabatically connected to the electron gas reference system, while sizable errors arise when the electronic configuration changes discontinuously upon gradual change of the external potential.

The reason for these trends can be traced to the interplay of correlation (and exchange) with the nodal structure of the ground state wave function,³⁶ which represents the basic difference between the two configurations crossing at $r_s=0.13$. A better description by DFT methods of the total energy on both sides of the transition is an important and challenging task, and provides a stringent test for new approximate schemes. Even more challenging appears to be the task of describing, in terms of the electron density alone, the jump in correlation energy at $r_s=0.13$, and the drastic change in its derivative with respect to the density.

Our results for the pair correlation functions provide useful information for a different family of approximations, based on the adiabatic connection,^{37,38} and relying on the modeling of the exchange-correlation hole. The pair correlation functions centered on highly inhomogeneous regions (such as the background edge) do not converge quickly to their asymptotic value of 1, and display large nonspherical components. It is possible that methods based on the exchange-correlation hole, aiming at an accurate description of both total energies and Kohn-Sham (exact) eigenvalues, will need to include both these features, since the long range in the pair distribution function is reflected in long range electron-electron interactions, while the nonspherical components, irrelevant for the computation of total energies, do affect the single-electron eigenvalues.

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